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Blobo Clone – Angry Birds Toy Upgrade

Requirement and implementation specification and test plan

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1. Introduction

1.1 Purpose

The purpose of this document is to specify what this device is about, what is wanted from it, and how it is implemented and tested.

Requirement specification defines the requirements for Blobo clone – Angry Birds Toy Upgrade and preliminary measurements or actions which enable to define the implementation of the first version of the device.

Implementation specification defines the implementation of Blobo clone – Angry Birds Toy and if required, tests that enable defining the implementation in sufficient detail. The implementation specification is based on the requirements of the requirement specification. It must give answers how each requirement becomes fulfilled. Implementation specification may contain several options for implementation. If possible, it would be good to estimate also the time required to complete each implementation.

Test plan defines the tests that are performed during and after the development process for the device. Tests must verify that the device meets the requirements.

1.2 Scope

Requirement specification covers the functional, physical and electrical requirements of the device.

Implementation specification covers the implementation techniques, protocols, components, PCB and software.

Test plan covers descriptions of basic hardware, unit, integration, system and regression tests.

1.3 Human resources

Project team includes Student1 and Student2. Both team members will be working in all project phases. The areas of responsibility are:

³⁵ ₁₇	electronics	Student1
³⁵ ₁₇	software	Student1
³⁵ ₁₇	integration	Student2
³⁵ ₁₇	documentation	Student2
³⁵ ₁₇	testing	
³⁵ ₁₇	project management	Student1

1.4 Acronyms and abbreviations

AD	Analog to Digital
GND	Ground line (negative supply)
MCU	Microcontroller Unit



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PC	Personal Computer
PCB	Printed Circuit Board
Vcc	Common-collector voltage a.k.a positive supply voltage or operating voltage
JTAG	Joint Test Action Group
EEPROM	Electrically Erasable Programmable Read-Only Memory
TTL	Transistor–transistor logic



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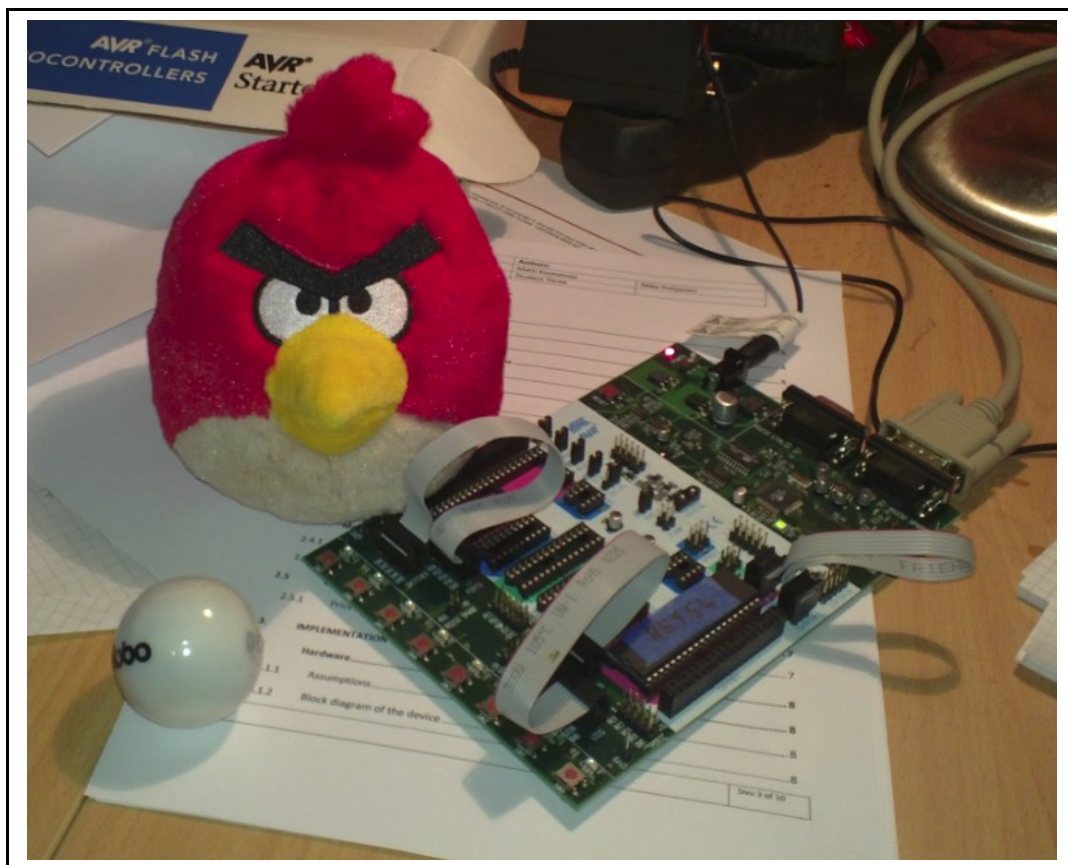
1.5 Overview

The purpose of this project is to upgrade existing Angry Birds soft toy with features of a Blobo device.

The toy has a push button and an audio speaker. When the push button is pressed a sound is played. There are three different audio sounds and they are played in a fixed order. Average length of audio sound is 2 seconds. There is unchangeable battery inside the toy.

The plan is to add more features to make the toy more usable and more fun. We will add Blobo device features such as 3-dimensional accelerator and orientation sensors, pressure sensor, RGB leds and bluetooth connectivity. The existing audio speaker and the push button are reused from the toy. The user interface will be redesigned to utilize the new features.

The device will have two modes, the play mode and the control mode. In the play mode the device will recognise gestures and react by changing the color of RGB leds and by playing audio sounds. In the control mode the device can be controlled using bluetooth or serial connection. The color of RGB leds can be changed and sounds can be triggered. Also the sensor data can be read in the control mode. At this first prototype the device is not used as a game controller like Blobo is used.



Picture 1 – Angry Birds soft toy, Blobo ball and SDK 500 development board



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2. Requirement specification

2.1 Assumptions

The toy will be used by children of small age so the safety of the actual product have to be taken care of. For the prototype device these safety issues are out of scope.

Due the copyright issues of the Angry Birds soft toy no plans for actual commercial production is planned. Only the prototype unit will be made.

2.1 Functional requirements

2.1.1 General functionality

Req-2.1.1	Device must have a two modes, the play mode and the control mode.
Req-2.1.2	Device must restore previous mode when it is turned on.
Req-2.1.3	Device must have a reset button (button R).
Req-2.1.4	The device must have a push button (button A).
Req-2.1.5	The device must have mode switch (switch A).
Req-2.1.6	The device must have two RGB leds. (led A, led B). Leds must be controllable separately.
Req-2.1.7	The device must detect 3-dimensional acceleration (min 0 G, max 2 G, resolution 0.1 G min).
Req-2.1.8	The device must detect 3-dimensional orientation (resolution 1 degree min).
Req-2.1.9	The device must detect pressure from some part of the casing.
Req-2.1.10	Device must be capable to play at least three different audio sounds (8bit PCM audio, 8KHz, min length 1 second).
Req-2.1.11	Device must have bluetooth connectivity. Serial Port Profile (SPP) must be supported and be used to detect commands and deliver sensor data. Device must have at least 10 m range for the bluetooth connectivity.
Req-2.1.12	Device must have status led for power. Led is on when power is connected and device is running.
Req-2.1.13	Device must have status led for bluetooth activity. Led is on when data is sent or received via bluetooth connection.
Req-2.1.14	Device must have status led for error. Led is on when an error is detected.
Req-2.1.15	Device must have a power switch.
Req-2.1.16	Device must have status led for the device mode. Led is on when the play mode is active.



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Opt-2.1.17	Device may have internal error log stored to EEPROM. The error log must have space to at least 10 last error. The log must have human readable error description and timestamp.
Opt-2.1.18	Device may have a socket for external clock crystal and switch for enabling it.
Opt-2.1.19	Device may have a control interface for vibration motor.

2.1.2 Play mode functionality

Req-2.2.1	Initial status of RGB leds is off when the device is in play mode.
Req-2.2.2	Device must recognize "push a button" gesture (button A) in the play mode. Response may include RGB led status change and audio sound.
Req-2.2.3	Device must recognize "squeeze the nose" gesture (pressure sensor) in the play mode. Response may include RGB led status change and audio sound.
Req-2.2.4	Device must recognize "zero gravity" gesture (when the device is thrown up in the air and it reaches the top altitude) in the play mode. Response may include RGB led status change and audio sound.
Req-2.2.5	Device must recognize "free fall" gesture (when the device dropped) in the play mode. Response may include RGB led status change and audio sound.
Req-2.2.6	Device must recognize "upside down" gesture (when the device is turned upside down) in the play mode. Response may include RGB led status change and audio sound.

2.1.3 Control mode functionality

In control mode the device reacts to serial data input sent via bluetooth connection. The data is sent as serial output via bluetooth connection.

Req-2.3.1	Initial status of RGB leds is blue when the device is in control mode.
Req-2.3.2	Acceleration sensor data (x,y,z) must be sent to bluetooth serial connection when command "ACC" is sent to device. Exact data format will be defined later.
Req-2.3.3	Pressure sensor data must be sent to bluetooth serial connection when command "PRE" is sent to device. Exact data format will be defined later.
Req-2.3.4	Push button status must be sent to bluetooth serial connection when command "BUT" is sent to device. Exact data format will be defined later.
Req-2.3.5	Color of RGB leds must be sent to bluetooth serial connection when command "LED" is sent to device. Exact data format will be defined later.
Req-2.3.6	Audio sound must be triggered when command "PLAY<sample id>" (<sample id> is id of the



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	audio clip, eg. PLAY2 plays audio sample number 2) is sent to device.
Req-2.3.7	Color of RGB leds must be set when command "LED<led id><RGB>" (<led id> is the id of the led, A or B, <RGB> is 1 for led on and 0 for led off, eg. LEDA110 sets red and green on and blue off for led A) is sent to device.

2.2 Electrical requirements

2.2.1 Operating voltage

Req-3.1	Device must use 4.5 – 5.5V operating voltage.
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2.2.2 Power source

Req-3.2	Device must use 9 V battery as a power source.
Req-3.3	The operating time must be at least 1 hour with full battery.

2.3 Mechanical requirements

2.3.1 Casing and size

Req-4.1	Device must use given Angry Birds soft toy as casing. If the casing of the toy is found out to be too small for the prototype device a larger toy can be used instead
Req-4.2	The device casing must have a zipper for changing the batteries.
Req-4.3	The two RGB leds must be used as the eyes of the toy.
Req-4.4	The push button must not be visible but still operatable from outside of the casing.
Req-4.5	The mode switch must not be visible or operatable from outside of the casing
Req-4.6	The pressure sensor must be not visible but still operatable from outside of the casing.

2.3.2 External connectors

Req-4.7	The device must have JTAG connector for debugging purposes.
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2.4 Other requirements

2.4.1 Price

Req-5.1	Price of a single component must not be higher than 25 euro.
Req-5.2	Total price of the prototype must not be higher than 100 euro.

2.5 Future features

These optional features are not planned to be implemented for the prototype. These are ideas for future development of the device.

Opt-6.1	The device may have audio microphone for audio input.
Opt -6.2	The device may have a vibrator for physical feedback.
Opt -6.3	The device may have sleep mode for saving battery life.
Opt -6.4	The device may have Blobo controller compability.
Opt -6.5	Mobile application may be development for controlling the device.
Opt -6.6	SD-card reader may be included for audio sound storage.
Opt-6.7	Continous sensor data delivery via bluetooth connection.
Opt-6.8	Sensors to detect nearby similar toys.



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3. Implementation specification

3.1 Hardware

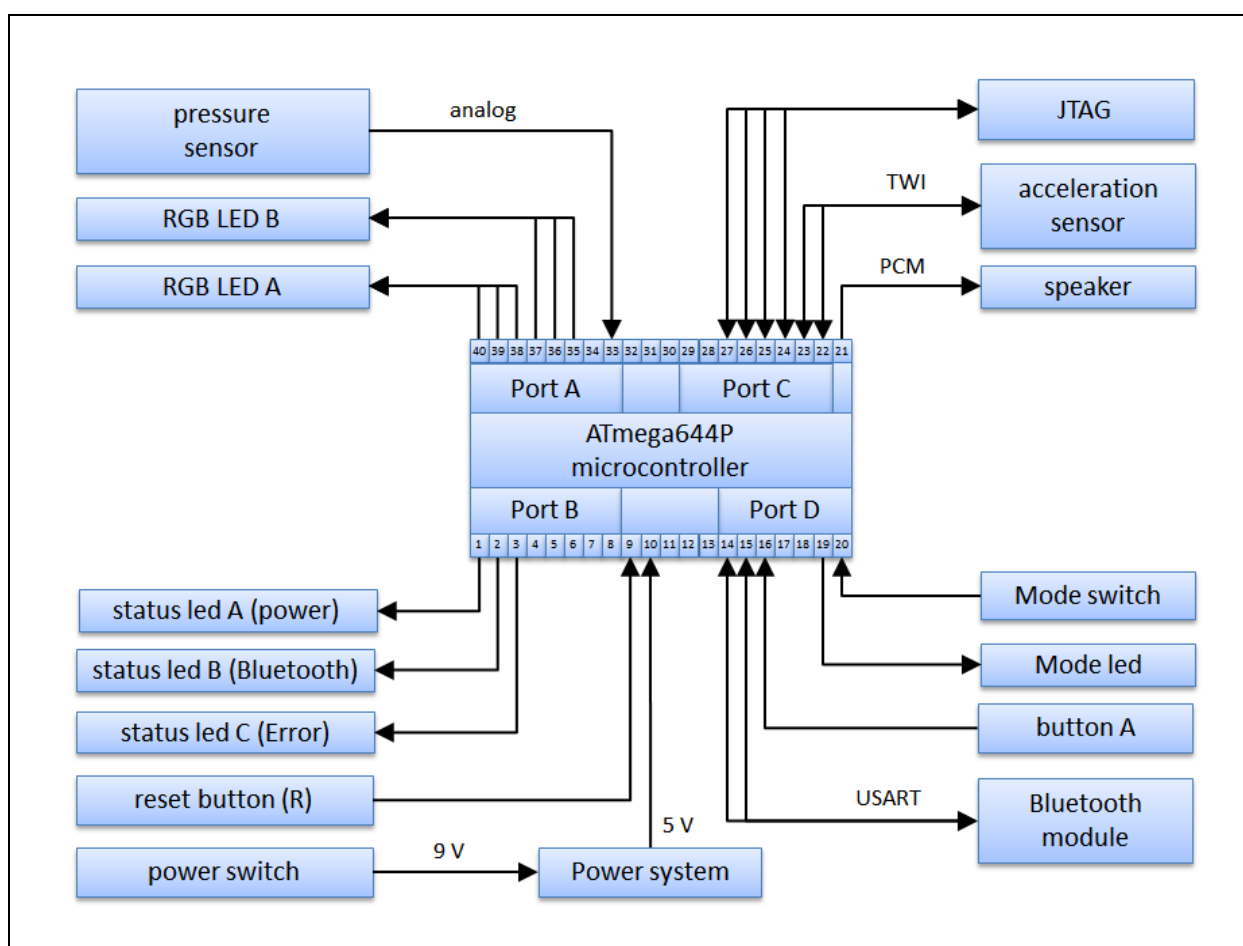
3.1.1 Assumptions

The development team uses STK500 development board and a breadboard for testing the components and prototyping the software. This prototyping with real components can be done simultaneously with actual PCB design and this should increase development speed significantly.

The development team purchase main components of the prototype device. The most expensive elements (MCU, bluetooth module, acceleration sensor module) are all under 20 euros each. (Req-5.1)

The total cost of the elements is kept under 100 euros. (Req-5.2)

3.1.2 Block diagram of the device



PICTURE: Block diagram of the device. The connection types and pin numbers of the MCU are shown.



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3.1.3 MCU

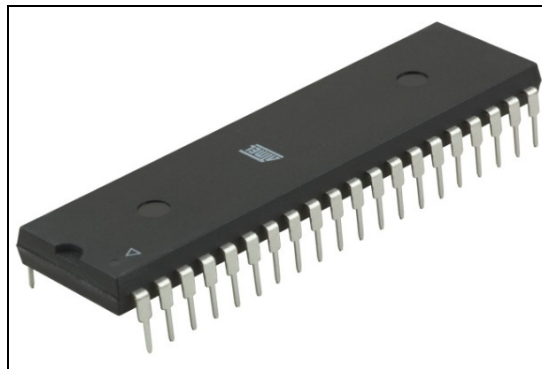
ATmega644P [1] microcontroller is used as MCU of the device. The operating frequency will be determined later during the testing phase. Initial assumption is to use the maximal internal clock (8MHz). A socket will be reserved for a external clock crystal.

Main reason for using Atmega64 over Atmega32 is the double flash memory size. The flash memory is used to store audio data. If more flash memory is needed or there are problems using the flash of the MCU then an option is to use external memory like SD card for the audio data.

Unused pins are connected to pin sockets for later use.

The MCU is inserted to a socket to be easy to replace.

If for some reason the selected MCU is unavailable then ATmega32 can be used instead.

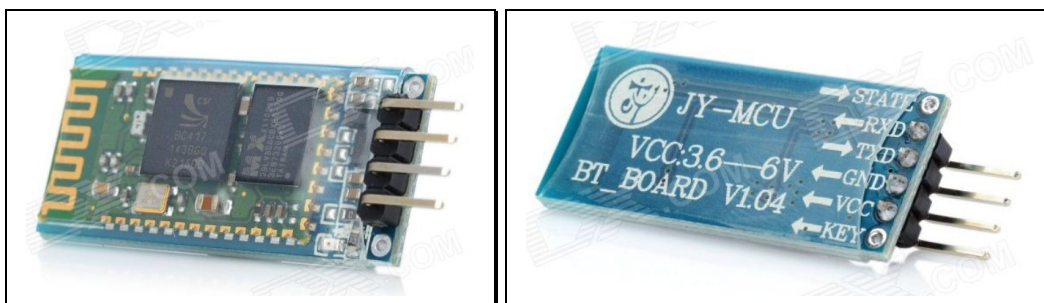


3.1.4 Bluetooth module

JY-MCU Arduino Bluetooth Wireless Serial Port Module [2] is used to provide Bluetooth connectivity. The Bluetooth module communicates with MCU using USART serial communication channel so it uses pins 14 and 15 of MCU. The connection parameters such as the baud rate are defined later during component testing phase. The module has an integrated antenna and it can use the same operation voltage (5 V) than the MCU.

The Bluetooth module is inserted into a socket for easy replacing.

(Req-2.1.11)



PICTURE: JY-MCU Bluetooth module front and back.



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3.1.5 Acceleration sensor module

ADXL345 acceleration sensor module [3] is selected because of its easy interface, operation voltage and price. The acceleration sensor module offers SPI and TWI interfaces for the acceleration data. It has two external interrupt lines which can be used to deliver state changes.

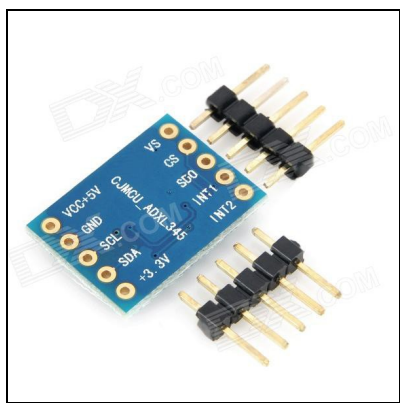
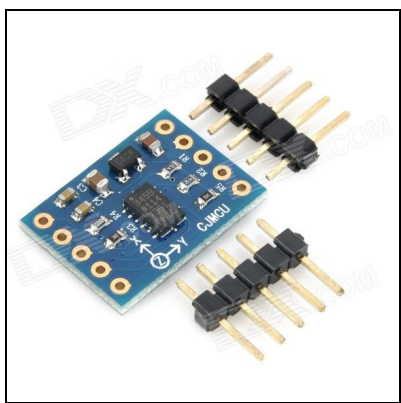
It is connected to MCU using TWI channel. It uses pins 22-23 of the MCU. Module offers digital acceleration data and device orientation can be calculated from this data by software. An interrupt is used to signal MCU about new data.

The operation voltage of the module is from 3.3V to 5.0 V so the same 5V operation voltage can be used with MCU.

If for some reason the selected acceleration module is not usable then an another acceleration sensor module with PWM output for each of the x,y and z channels is used. If this option is selected then pins 33-36 of the MCU are used with ADC to get the acceleration data. In this case pins 4 and 17 are used to control RGB leds instead.

The Acceleration sensor module is inserted into a socket for easy replacing.

(Req-2.1.7)



PICTURE: ADXL345 accceleration sensor front and back.

3.1.6 Pressure sensor module

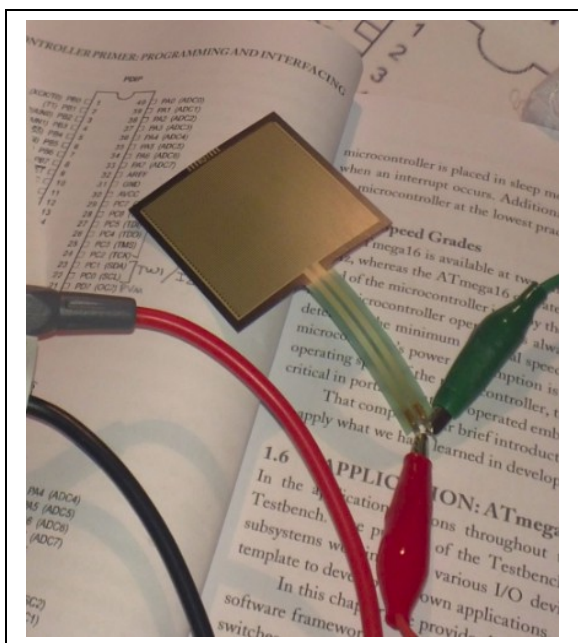
The analog pressure sensor changes the internal resistance according to the pressure agains its surface. The pressure sensor is connected to pin 33 of the MCU. The analog sensor output is converted to digital by MCU ADC system. (Req-2.1.9)

The sensor module is inserted inside the nose of the soft toy. The nose is made from soft material so the pressure sensor can be operated through the surface of the toy. (Req-4.6)

The pressure sensor is connected to PCB with removable cable for easy replacing.



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PICTURE: The pressure sensor with some testing cables attacheded.

3.1.7 Audio speaker

Existing audio speaker of the soft toy is reused to playback audio sounds. 40KB of FLASH memory of MCU is reserved for 5 second of audio data. Timer 2 is used with interrupt to read audio samples and control PCM output of pin 21 of the MCU. Timer 0 is used to procude PCM output to control the speaker element. (Req-2.1.10)

If the volume of the speaker is not enough a simple amplifier will be used to procude necessary audio volume.

If the existing speaker element is not usable then an alternative speaker is used.

The audio speaker is connected to PCB with removable cable for easy replacing.

3.1.8 RGB leds

The two rgb leds are controlled using pins 35-40 (6 pins) of MCU. Each control pin will have resistor to reduce the operation current of the leds. (Req-2.1.6)

Two RGB leds will be inserted as eyes of the soft toy and connected with wires to the PCB. Some kind of support structure must be added to the soft toy to offer stable platform for the leds. (Req-4.3)

The RGB led board is connected to PCB with removable cable for easy replacing.



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3.1.9 Status leds

The three status leds are connected to pins 1-3 of the MCU. Resistors are used to keep current at decent level. (Req-2.1.12, Req-2.1.13, Req-2.1.14)

3.1.10 Mode led

The mode led is connected to pin 19 of the MCU. A resistor is used to keep current at decent level. (Req-2.1.16)

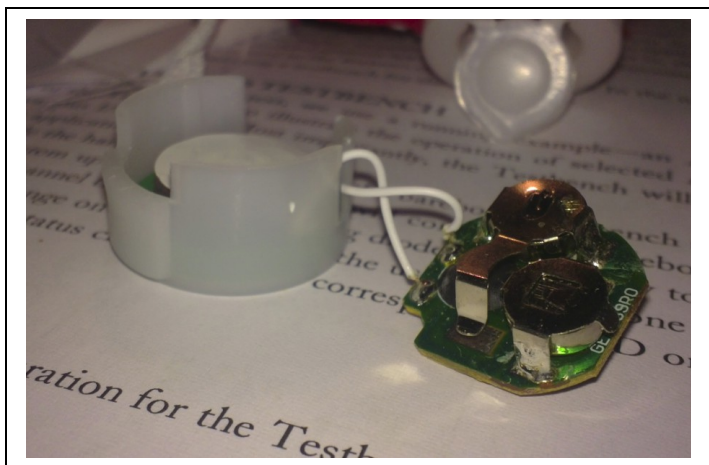
3.1.11 Switch A (mode switch)

Mode switch is connected to pin 20 of the MCU. (Req-2.1.5). A dip switch is placed to PCB and cannot be operated when the zipper is closed. (Req-4.5)

3.1.12 Button A

Button A is connected to external interrupt pin 16 (INT0) of the MCU. (Req-2.1.4)

The button is inserted inside the original HW case and put in the original position. So the physical interface remains similar to the original product. (Req-4.4)



PICTURE: The hardware of the original toy. The audio speaker is on the left and the main electronics with two batteries and mechanical button on the right.



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3.1.13 Reset button (button R)

Reset button is connected to reset pin 9 (RESET) of the MCU. The MCU internal reset logic handles the reset. (Req-2.1.3).

3.1.14 Power switch

Power switch is connected to battery in and it will cut the positive(+) and negative(-) lines from battery. (Req-2.1.15).

3.1.15 JTAG connector

Pins 24-27 (4 pins) are connected to pin headers(2x5) for the JTAG connector. (Req-4.7)

Optionally the pin header (2x5) can be connected to PCB using a wire so it is easier to use when the PCB is inserted inside the case.

3.1.16 PCB

The PCB board must be round shaped with 75 mm diameter to fit inside the soft toy casing. The PCB size might become a problem for layout design.

Optional sockets are integrated to design so it is easier to make new features with software later.

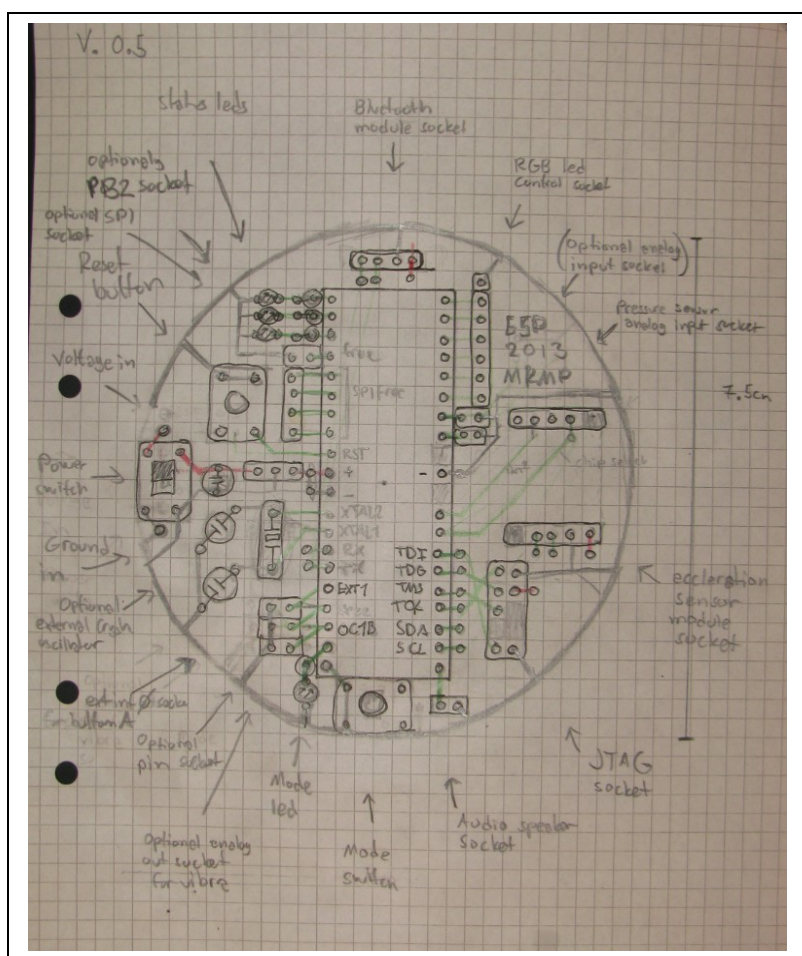
Some lines like JTAG and USART data lines are routed using jumper lines. Also operation voltage is provided to the components using jumper lines. The lines are routed under the PCB.

The device is designed to be flat to fit inside the casing. The acceleration module is placed next to the MCU and the bluetooth module is turned 90 degrees to be placed directly above and in line with the MCU.

A separate PCB is designed for the RGB leds. This can be produced on the same PCB with the main board and removed later by sawing.



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PICTURE: The initial sketch of the PCB.

3.1.17 Power

9 V battery is used with 7805 regulator to produce 5.0 V operation voltage. Additional capacitors are used to filter power current. Battery connector will be added to main board of the device. (Req-2.3.1, Req-2.3.2)

A typical alkaline 9V battery has around 550 mAh capacity. Maximal total current of the device must be below 550 mA. So the maximal power consumption of the device is 4.95 W per hour.

Here is listed the main components with their typical power consumptions:

³⁵ ₁₇	ATmega644P MCU	1 mA, 5 V = 5mW
³⁵ ₁₇	Bluetooth module	1 mA, 5V = 5 mW
³⁵ ₁₇	Accelerator sensor	1 mA, 5V = 5 mW
³⁵ ₁₇	Two RGB leds	120 mA, 5V = 500mW
³⁵ ₁₇	Three status leds	40 mA, 5V = 200mW



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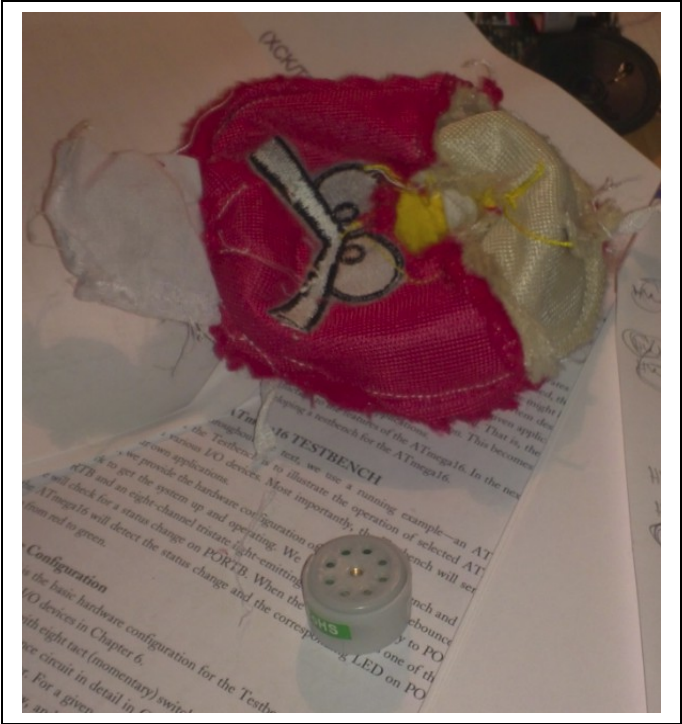
³⁵ ₁₇	Mode led	20 mA, 5V = 100 mW
³⁵ ₁₇	Voltage regulator	203 mA, 4V voltage drop, 812 mW
³⁵ ₁₇	Other components	20 mA (raw estimation), 5V = 100 mW

Total power consumption in normal use is around 1.727 W which is lower than maximal 4.95 W .
(Req.2.3.3)

3.1.18 Casing

The soft toy is opened and used as the casing of the device. PCB will have separate case inside the soft toy.
(Req-4.1)

A zipper will be added to the casing of the soft toy. (Req-4.2)



PICTURE: The casing of the original toy (turned inside out) and the hardware casing of the original toy.

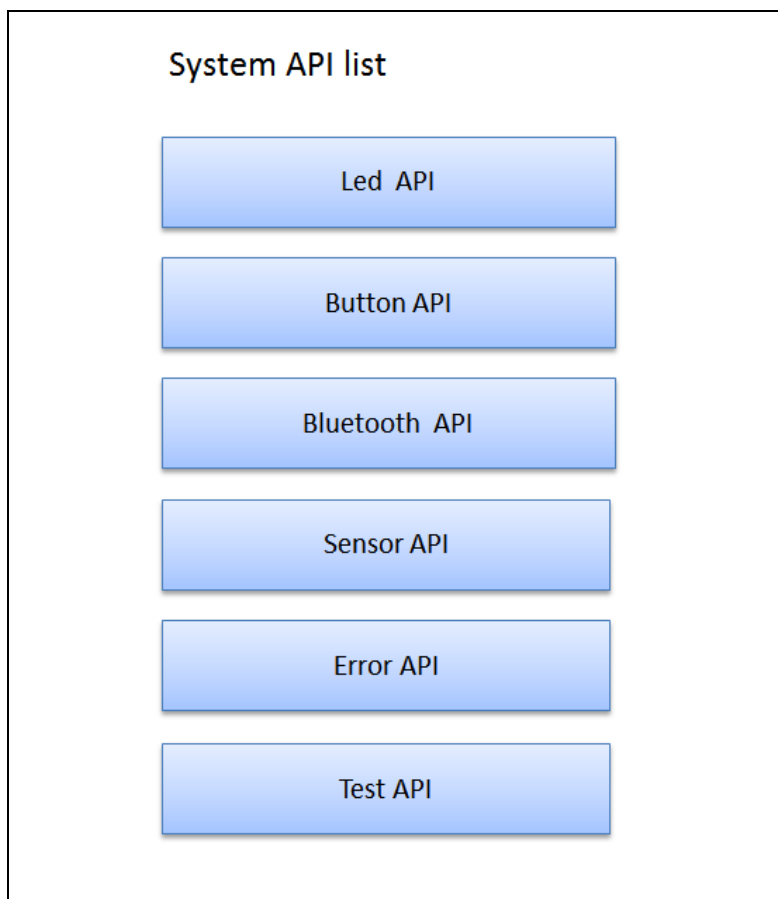


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3.2 Software

3.2.1 Software architecture

Functionality is divided into separate modules and each module offers easy to use API for the rest of the software.



PICTURE: The API modules of the system.



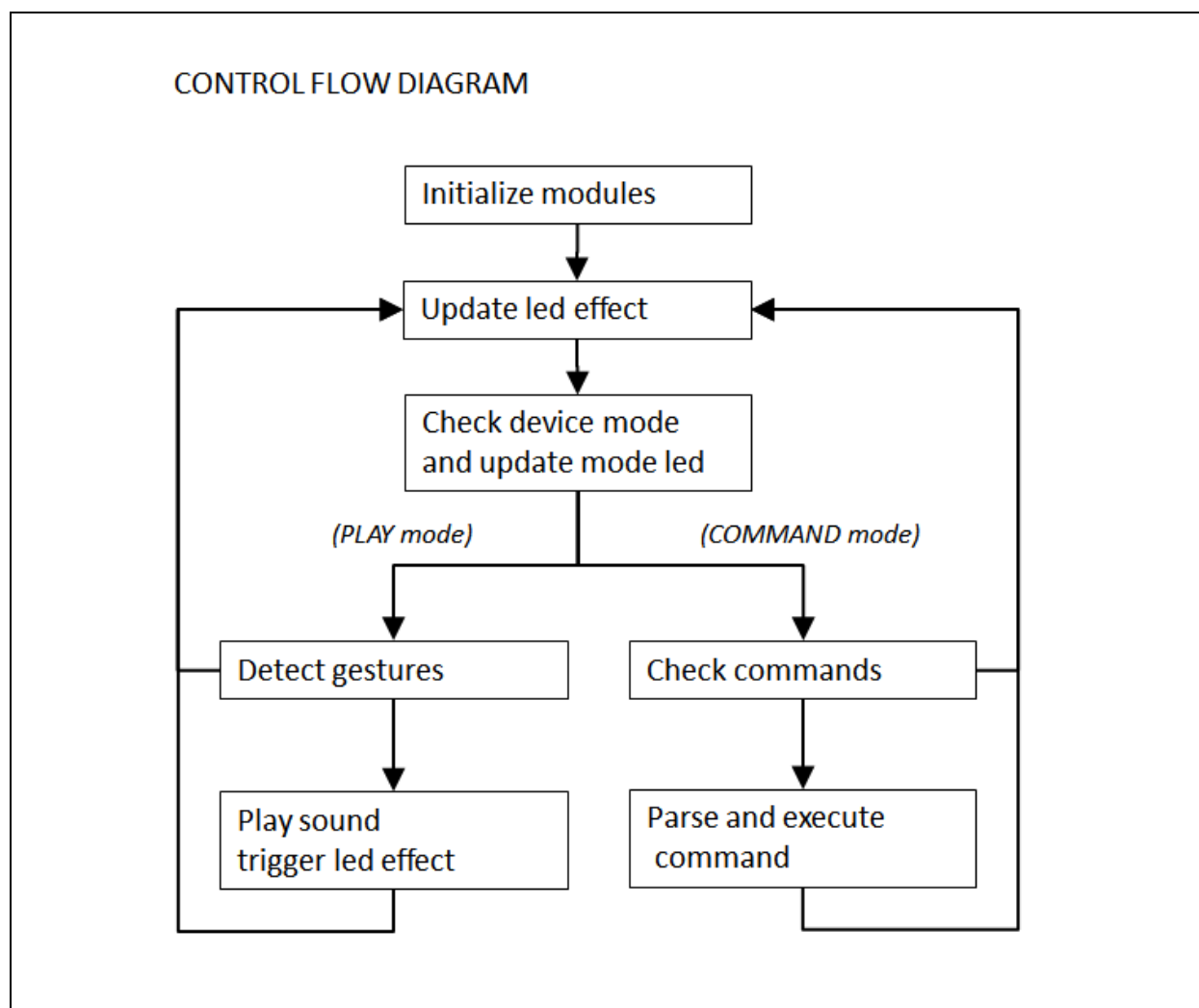
	Authors:	
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3.2.2 Initialization and Main Loop

When the device is turned on the previous device mode (play/control) is read from EEPROM and written to global variable (Req-2.1.1, Req-2.1.2).

After the device mode is read then the rest of the initialization is made and the main loop is started.

Power status led is set on at the beginning of the software main function. (Req-2.1.12)



PICTURE: The control flow diagram of the system.

3.2.3 Error handling

Error handling API is defined to be used by the rest of the software. When error is reported the error status led is set on and error text is stored to EEPROM. All modules report their internal errors using error reporting API. (Req-2.1.14, Opt-2.1.17)

Optionally the main loop running can be monitored by setting a timer in the beginning of the main loop. If main loop is never started or it stalls the timer gets triggered and error is reported.



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3.2.4 Button A

Button A is connected to external interrupt pin of the MCU. Button pressed flag is set in Interrupt routine of the interrupt. Button A API is defined to be used by the rest of the software. (Req-2.1.4).

3.2.5 Switch A (mode switch)

Mode switch is connected to pin 20 of the MCU. The state of the pin 20 is read inside the main loop and global variable is updated. (Req-2.1.5).

Another way is to set pin state changed interrupt for the pin.

Button A API is defined to be used by the rest of the software.

3.2.6 Led control in play mode

RGB leds are controlled using port A of the MCU. A led control API is defined to be used by the rest of the software to manipulate and query led statuses. (Req-2.1.6)

Initial status of the led is before entering to main loop. (Req-2.2.1, Req-2.3.1)

For play mode gesture responses a set of special effects is defined. These effects can be triggered from anywhere in the software and are implemented in main loop by changing led statuses. Effects are simple like blinking one color etc.

(Req-2.2.2, Req-2.2.3, Req-2.2.4, Req-2.2.5, Req-2.2.6).

Mode led is set on when play mode is active (Req-2.1.16).

3.2.7 Led control in control mode

If led status changing command is received from bluetooth serial connection in control mode then the led status is changed.

When led related command is received from bluetooth serial connection it is handled by UART data ready interrupt handling routine. If the command is "LED" then the statuses of each led is written to output buffer of the bluetooth serial data connection. If the command is LED<led id><RGB> then the <led id> and <RGB> is parsed and the led status are updated.

(Req-2.1.6, Req-2.3.5, Req-2.3.7)

Bluetooth communication status led is set high when in or out buffer is not empty. (Req-2.1.17)



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3.2.8 Audio playback

Audio data is stored in the flash memory of the MCU. Audio clip with 8 bit 8Khz PWM takes 8000 bytes/s. The system will have 5 audio clips 1 second each. Total memory for the audio data is 40 Kbytes.

Audio playback API is defined and used by the rest of the software. There is function for start audio playback with given audio sample and a function to stop audio playback.

Timer, interrupt and PWM systems of the MCU are used for actual playback. A timer is activated when playback is on and an interrupt vector is called to read audio data and control PWM output of pin 21 of the MCU.

(Req-2.1.10)

3.2.9 Acceleration sensor data reading

The acceleration sensor data is read from the sensor module using TWI channel. There are two ways to read the data from the registers of the sensor module. We can connect data ready signal of the sensor module to an external interrupt pin of the MCU and read when this interrupt is triggered. An another way is to poll the sensor data in the main loop or in timer triggered function. The proper way is selected later after the module is tested with STK500.

In spite of the reading method the latest acceleration data is written to global variables (3 x 16 bits = 6 bytes) and acceleration sensor API is defined to be used by the rest of the software. (Req-2.1.7)

The orientation of the device is calculated from the acceleration vectors (x,y,z). (Req-2.1.8)

3.2.10 Pressure sensor

An analog pressure sensor is connected to pin 33 of the MCU. Analog to Digital conversion (ADC) feature of the MCU is used to get digital input. Pressure sensor API is defined to be used by the rest of the software. (Req-2.1.9)

3.2.11 Bluetooth serial connection

TTL level USART connection is used to communicate with bluetooth module. USART is initialized before entering to main loop. Interrupt handler is used to read serial input. The data is stored to an input buffer. When carriage return character (ascii 13) is received a command counter is updated.

The response data is stored to separate output buffer and interrupt vector is used to send the data from this buffer to the USART.

Before the bluetooth module is operational the system logic is tested using R232 connection of the STK-500 development board.



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Bluetooth connectivity API is defined to be used by the rest of the software.

The bluetooth module implements Serial Port Profile (SPP)

(Req-2.1.11)

In the main loop the command counter is checked and if commands are received they are parsed and executed.

- ³⁵₁₇ When "ACC" command is received the acceleration sensor data is read from global variable and written to output buffer and transmission is started. (Req-2.3.2)
- ³⁵₁₇ When "PRE" command is received the pressure sensor data is read from global variable and written to output buffer and transmission is started (Req-2.3.3)
- ³⁵₁₇ When "BUT" command is received the state of the button A is read from global variable and written to output buffer and transmission is started. (Req-2.3.4)
- ³⁵₁₇ When "LED" command is received the statuses of the leds are read from global variable and written to output buffer and transmission is started. (Req-2.3.5)
- ³⁵₁₇ When "PLAY<sound id>" command is received the <sound id> is parsed and selected audio sample playing is started. (Req-2.3.6)
- ³⁵₁₇ When "LED<led id><RGB>" command is received the <led id> and <RGB> are parsed and the led statuses are changed. (Req-2.3.7)

3.2.12 Gesture recognition and feedback

Gestures are recognized in the main loop by testing the sensor values.

- ³⁵₁₇ A "Push a button" gesture is recognized using button API. When the gesture is recognized the led effect and audio playback are triggered in the play mode. (Req-2.2.2)
- ³⁵₁₇ A "squeeze the nose" gesture is recognized by using pressure sensor API.. When the gesture is recognized the led effect and audio playback are triggered in the play mode. (Req-2.2.3)
- ³⁵₁₇ A "zero gravity" gesture is recognized by checking the threshold value of combined acceleration vector when the global variables are updated. Optionally the zero gravity detection system of the acceleration sensor can be used. When the gesture is recognized the led effect and audio playback are triggered in the play mode. (Req-2.2.4)

The "freefall" and "upside down" gestures are disabled for a couple of seconds to avoid gesture overlapping.



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³⁵₁₇ A "free fall" gesture is recognized by checking the treshold value of combined acceleration vector when the global variables are updated. Optionally the free fall detection system of the acceleration sensor can be used. When the gesture is recognized the led effect and audio playback are triggered in the play mode. (Req-2.2.5)

The "upside down" gesture is disabled for a couple of seconds to avoid gesture overlapping.

³⁵₁₇ A "upside down" gesture is recognized by checking the treshold value of combined acceleration vector when the global variables are updated. Optionally the free fall detection system of the acceleration sensor can be used. When the gesture is recognized the led effect and audio playback are triggered in the play mode. (Req-2.2.6)



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3.3 Cost calculations

3.3.1 Components

³⁵ ₁₇	ATMega64	16 eur
³⁵ ₁₇	Bluetooth module	8 eur
³⁵ ₁₇	Acceleration sensor module	8 eur
³⁵ ₁₇	Pressure sensor	2 eur
³⁵ ₁₇	RGB leds	2 eur
³⁵ ₁₇	PCB	(free)
³⁵ ₁₇	Small general components	15 eur
³⁵ ₁₇	Angry Birds soft toy casing	15 eur
Total		66 eur

3.3.2 Development tools

³⁵ ₁₇	STK500 development board	150 eur
³⁵ ₁₇	Voltage meter	70 eur
³⁵ ₁₇	Small components and tools	50 eur
³⁵ ₁₇	JTAGICE3 debug tool	100 eur
Total		370 eur

3.3.3 Total costs

Total project cost is estimated to be around 436 euros.



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3.4 Tools

Eagle application will be used for schematics and PCB layout design. Atmel Studio 6.0 will be used for software development. STK 500 development board with a breadboard will be used for prototyping both hardware and software implementation before the prototype HW is available.

Also a JTAGICE3 debugging tool is used to speed up out of the office/university product development.



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4. TESTING

4.1 Basic hardware tests without power and without ICs on board

Test ID	Description	Result
Test-1.1.1	Before connecting IC:s and power	
Test-1.1.2	Visual inspection shows no short cuts or bad solder connections.	
Test-1.1.3	GND and Vcc lines are not short cutted.	
Test-1.1.4	GND is in contact from DC connector to the GND pins of IC sockets.	
Test-1.1.5	Vcc line is in contact from DC connector to the Vcc pins of IC sockets.	
Test-1.1.6	GND and Vcc of the JTAG connector are in contact to GND and Vcc nets of the PCB	
Test-1.1.7	Data pins of the JTAG connector are in contact to corresponding pins in MCU	
Test-1.1.8	Data pins of the JTAG connector are not in contact to GND or Vcc	
Test-1.1.9	Data pins of the bluetooth module are in contact to corresponding pins in MCU	
Test-1.1.10	Data pins of the bluetooth module are not in contact to GND or Vcc	
Test-1.1.11	Data pins of the acceleration sensor module are in contact to corresponding pins in MCU	
Test-1.1.12	Data pins of the acceleration sensor module are not in contact to GND or Vcc	
Test-1.1.13	Button A is connected to ground and to reset pin of the MCU.	
Test-1.1.14	Speaker connector has one pin connected to the ground and another pin connected to correct output pin of MCU.	
Test-1.1.15	All the leds have proper resistors connected in serial with the led.	
Test-1.1.16	Pressure sensor connector has one pin connected to ground and another pin to correct input pin of the MCU	
Test-1.1.17	RGB led connector has 6 pins connected to correct pins of MCU.	
Test-1.1.18	RGB led ground connector is connected to main board ground.	



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4.2 Basic hardware tests with power and without ICs on board

Test ID	Description	Result
Test-1.2.1	When powering on the device the current consumption is under 50 mA	
Test-1.2.2	Voltage between Vcc and GND is 5 volts within tolerance of 0.2 V	
Test-1.2.3	Any component does not heat up rapidly	

4.3 Tests after connecting IC:s without power

Test ID	Description	Result
Test-1.3.1	GND and Vcc are not in short circuit	

4.4 Tests when connecting power to PCB with IC:s

Test ID	Description	Result
Test-1.4.1	Any component does not heat up rapidly.	
Test-1.4.2	The current consumption is under 100 mA.	



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4.5 Unit tests

Test-2.1	MCU is operational
Tools	Device with microcontroller attached.
Steps	Switch device power on.
Accepted result	The green power led is lit.
Test result	

Test-2.2	Bluetooth module is operational
Tools	Device with microcontroller and bluetooth module attached. A computer with bluetooth connectivity. Terminal application (eg. Putty).
Steps	Switch device power on. Establish bluetooth connection to the device. Send "test bluetooth" command to the device.
Accepted result	Device must send response: "bluetooth test OK."
Test result	

Test-2.3	Acceleration sensor module is operational
Tools	Device with microcontroller with RGB led unit and acceleration sensor module attached.
Steps	Enable acceleration sensor test from source code. Flash new software to the device. Hold device in normal position. Switch device power on. Turn device upside down. After test flash the original software back to the device.
Accepted result	When device is at normal position the RGB led must be red. When the device is turned upside down the RGB led must be blue.
Test result	

Test-2.3	Pressure sensor is operational
Tools	Device with microcontroller with RGB led unit and pressure sensor attached.
Steps	Enable pressure sensor test from source code. Flash new software to the device. Switch device power on. Press the pressure sensor element. After test flash the



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	original software back to the device.
Accepted result	When pressure sensor element is not pressed the RGB led must be red. When the pressure sensor element is pressed the RGB led must be blue.
Test result	

Test-2.4	Audio playback is operational
Tools	Device with microcontroller and pressure sensor attached.
Steps	Enable audio playback test from source code. Flash new software to the device. Switch device power on. After test flash the original software back to the device.
Accepted result	When device is turned on a audio sound is played.
Test result	

Test-2.5	Leds are operational
Tools	Device with microcontroller with RGB led unit attached.
Steps	Enable led test from source code. Flash new software to the device. Switch device power on. After test flash the original software back to the device.
Accepted result	When device is turned on all the leds should be blinking.
Test result	

Test-2.6	Button A is operational
Tools	Device with microcontroller with button A and RGB led unit attached.
Steps	Enable button test from source code. Flash new software to the device. Switch device power on. Press button A. After test flash the original software back to the device.
Accepted result	When device is turned on RGB leds must be red. After the button A is pressed the RGB leds must be blue.
Test result	

Test-2.7	Mode switch is operational
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Tools	Device with microcontroller.
Steps	Enable button test from source code. Flash new software to the device. Switch device power on. Set mode switch on. Set mode switch off. After test flash the original software back to the device.
Accepted result	When device is turned on and mode switch is on the mode led should be on. When the mode switch is off the mode led should be off.
Test result	

Test-2.8	Reset button is operational
Tools	Device with microcontroller.
Steps	Enable reset button test from source code. Flash new software to the device. Switch device power on. Press the reset button. After test flash the original software back to the device.
Accepted result	When device is turned on the power led should become on after one second delay. When reset button is pressed the power led should get off and become on after one second.
Test result	

Test-2.9	Power switch is operational
Tools	Device with microcontroller. Voltage meter.
Steps	Turn device off. Measure voltage between pins 10 and 11. Turn device on. Measure voltage between pins 10 and 11.
Accepted result	When device is turned off the voltage must be 0 V. When device is turned on the voltage must be 5 V (within 0.2 V tolerance)
Test result	

Test-2.10	JTAG socket is operational
Tools	Device with microcontroller and JTAG debugging tool attached to JTAG socket. AtmelStudio application.
Steps	Enable JTAG test from the source code. Turn device on. Flash the device using JTAG connection. After test flash the original software back to the device.



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Accepted result	When the device is restarted after flashing JTAG test enabled software the power led should be blinking.
Test result	

Test-2.11	Power unit is operational
Tools	Device with microcontroller. Voltage meter.
Steps	Turn the device on. Measure voltage from battery in pins. Measure voltage from pins 10 and 10.
Accepted result	Voltage from battery in must be between 8V and 9 V. Voltage from the pins 10 and 11 must be 5V (within 0.2 V tolerance).
Test result	

Test-2.12	Casing is usable
Tools	Device with soft toy casing and main unit casing.
Steps	Put the main unit casing inside the soft toy casing and close the zipper.
Accepted result	The main unit casing fits inside the soft toy with zipper closed.
Test result	

4.6 Integration tests

These are defined later.

4.7 System tests

4.7.1 Tests for general functionality

Test ID	Requirement	Test description	Result
Test-4.1.1	Req-2.1.1	Press mode switch and ensure that mode led is on in play mode Test is passed if mode changes	
Test-4.1.2	Req-2.1.2	Change mode, turn off and turn on Test is passed if previous mode is on	



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Test-4.1.3	Req-2.1.3	Find reset button Test is passed if reset button is found	
Test-4.1.4	Req-2.1.4	Find push button Test is passed if push button is found	
Test-4.1.5	Req-2.1.5	Find mode switch Test is passed if mode switch is found	
Test-4.1.6	Req-2.1.6	Find two leds, control them in control mode Test is passed if two leds are found and controlled	
Test-4.1.7	Req-2.1.7	Move the toy in play mode Test is passed if a response is heard or seen	
Test-4.1.8	Req-2.1.8	Tilt the toy in play mode Test is passed if a response is heard or seen	
Test-4.1.9	Req-2.1.9	Press the toy in play mode Test is passed if a response is heard or seen	
Test-4.1.10	Req-2.1.10	Play three sounds in control mode Test is passed if at least three sounds are played	
Test-4.1.11	Req-2.1.11	Connect via bluetooth in control mode Test is passed if a reaction is received	
Test-4.1.12	Req-2.1.12	Turn on the toy Test is passed if power status led is blinking	
Test-4.1.13	Req-2.1.13	Send data via bluetooth Test is passed if bluetooth status led is lit	
Test-4.1.14	Req-2.1.14	Take off bluetooth module and turn on the toy Test is passed if error status led is lit	
Test-4.1.15	Req-2.1.15	Turn on the toy Test is passed if power is on	
Test-4.1.16	Req-2.1.16	Switch to play mode	



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		Test is passed if mode status led is lit Switch to control mode Test is passed if mode status led is not lit	
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4.7.2 Tests for play mode functionality

Test-4.2.1	Req-2.2.1	Switch to play mode Test is passed if play mode turns on and mode status led is lit	
Test-4.2.2	Req-2.2.2	Push push button Test is passed if a response is heard or seen	
Test-4.2.3	Req-2.2.3	Squeeze the nose Test is passed if a response is heard or seen	
Test-4.2.4	Req-2.2.4	Throw the toy up Test is passed if a response is heard or seen	
Test-4.2.5	Req-2.2.5	Drop the toy Test is passed if a response is heard or seen	
Test-4.2.6	Req-2.2.6	Put the toy upside down Test is passed if a response is heard or seen	

4.7.3 Tests for control mode functionality

Test-4.3.1	Req-2.3.1	Switch to control mode Test is passed if control mode turns on and mode status led is not lit	
Test-4.3.2	Req-2.3.2	Switch to control mode and send command "ACC" via bluetooth Test is passed if acceleration sensor data is received	
Test-4.3.3	Req-2.3.3	Switch to control mode and send command "PRE" via bluetooth	



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		Test is passed if pressure sensor data is received	
Test-4.3.4	Req-2.3.4	Switch to control mode and send command "BUT" via bluetooth Test is passed if button status is received	
Test-4.3.5	Req-2.3.5	Switch to control mode and send command "LED" via bluetooth Test is passed if color of RGB leds is received	
Test-4.3.6	Req-2.3.6	Switch to control mode and send command "PLAY[1,2,3]" via bluetooth Test is passed if sound is played	
Test-4.3.7	Req-2.3.7	Switch to control mode and send command "LED[A,B][0,1][0,1][0,1]" via bluetooth Test is passed if leds are lit as commanded	

4.7.4 Tests for mechanical requirements

Test-4.4.1	Req-4.1	Verify that casing is a Angry birds soft toy Test is passed if casing is a Angry bird soft toy	
Test-4.4.2	Req-4.2	Find zipper Test is passed if zipper is found	
Test-4.4.3	Req-4.3	Find two leds Test is passed if two leds are found in the eyes of the toy	
Test-4.4.4	Req-4.4	Operate push button without seeing it Test is passed if push button is pushed without seeing it	
Test-4.4.5	Req-4.5	Try to find mode switch Test is passed if mode switch is found inside the casing	
Test-4.4.6	Req-4.6	Operate pressure sensor without seeing it Test is passed if pressure sensor is operated but not seen outside the casing	
Test-4.4.7	Req-4.7	Find JTAG connector Test is passed if JTAG connector is found inside the casing	



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4.7.5 Tests for other requirements

Test-4.5.1	Req-5.1	Find out component prices Test is passed if none of the components pays more than 25 euros	
Test-4.5.2	Req-5.2	Count the total cost of the prototype Test is passed if the total cost is less than 100 euros	

4.8 Regression tests

These are defined later.



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